

# AWS Certified Solutions Architect - Associate SAA-C03

question 1001-1019

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## 【題號】question 1001

### 【題目】

A company operates a food delivery service. Because of recent growth, the company's order processing system is experiencing scaling problems during peak traffic hours. The current architecture includes Amazon EC2 instances in an Auto Scaling group that collect orders from an application. A second group of EC2 instances in an Auto Scaling group fulfills the orders.

The order collection process occurs quickly, but the order fulfillment process can take longer. Data must not be lost because of a scaling event.

A solutions architect must ensure that the order collection process and the order fulfillment process can both scale adequately during peak traffic hours.

Which solution will meet these requirements?

一家公司經營送餐服務。由於最近的成長，該公司的訂單處理系統在高峰流量時段遇到了擴展問題。目前架構包括 Auto Scaling 群組中的 Amazon EC2 實例，用於從應用程式收集訂單。Auto Scaling 組中的第二組 EC2 執行個體履行訂單。

訂單收集過程發生得很快，但訂單履行過程可能需要更長的時間。數據不得因縮放事件而遺失。

解決方案架構師必須確保訂單收集流程和訂單履行流程都能在尖峰流量時段充分擴展。

哪種解決方案可以滿足這些要求？

### 【選項】

A. Use Amazon CloudWatch to monitor the CPU Utilization metric for each instance in both Auto Scaling groups. Configure each Auto Scaling group's minimum capacity to meet its peak workload value.

B. Use Amazon CloudWatch to monitor the CPU Utilization metric for each instance in both Auto Scaling groups. Configure a CloudWatch alarm to invoke an Amazon Simple Notification Service (Amazon SNS) topic to create additional Auto Scaling groups on demand.

C. Provision two Amazon Simple Queue Service (Amazon SQS) queues. Use one SQS queue for order collection. Use the second SQS queue for order fulfillment. Configure the EC2 instances to poll their respective queues. Scale the Auto Scaling groups based on notifications that the queues send.

D. Provision two Amazon Simple Queue Service (Amazon SQS) queues. Use one SQS queue for order collection. Use the second SQS queue for order fulfillment. Configure the EC2 instances to poll their respective queues. Scale the Auto Scaling groups based on the number of messages in each queue.

【參考答案】D

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## 【題號】question 1002

### 【題目】

An online gaming company is transitioning user data storage to Amazon DynamoDB to support the company's growing user base. The current architecture includes DynamoDB tables that contain user profiles, achievements, and in-game transactions.

The company needs to design a robust, continuously available, and resilient DynamoDB architecture to maintain a seamless gaming experience for users.

Which solution will meet these requirements MOST cost-effectively?

一家線上遊戲公司正在將用戶資料儲存遷移到 Amazon DynamoDB，以支援公司不斷成長的用戶群。目前架構包括 DynamoDB 表，其中包含使用者個人資料、成就和遊戲內交易。

該公司需要設計一個強大、持續可用且有彈性的 DynamoDB 架構，以維持用戶的無縫遊戲體驗。

哪一種解決方案能夠最具成本效益地滿足這些要求？

### 【選項】

A.Create DynamoDB tables in a single AWS Region. Use on-demand capacity mode. Use global tables to replicate data across multiple Regions.

B.Use DynamoDB Accelerator (DAX) to cache frequently accessed data. Deploy tables in a single AWS Region and enable auto scaling. Configure Cross-Region Replication manually to additional Regions.

C.Create DynamoDB tables in multiple AWS Regions. Use on-demand capacity mode. Use DynamoDB Streams for Cross-Region Replication between Regions.

D.Use DynamoDB global tables for automatic multi-Region replication. Deploy tables in multiple AWS Regions. Use provisioned capacity mode. Enable auto scaling.

【參考答案】D

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## 【題號】question 1003

### 【題目】

A company runs its media rendering application on premises. The company wants to reduce storage costs and has moved all data to Amazon S3. The on-premises rendering application needs low-latency access to storage.

The company needs to design a storage solution for the application. The storage solution must maintain the desired application performance.

Which storage solution will meet these requirements in the MOST cost-effective way?

一家公司在本地運行其媒體渲染應用程式。該公司希望降低儲存成本，並將所有資料轉移到 Amazon S3。本地渲染應用程式需要低延遲地存取儲存。

該公司需要為該應用程式設計一個儲存解決方案。儲存解決方案必須保持所需的應用程式效能。

哪種儲存解決方案能夠以最具成本效益的方式滿足這些要求？

### 【選項】

A.Use Mountpoint for Amazon S3 to access the data in Amazon S3 for the on-premises application.

B. Configure an Amazon S3 File Gateway to provide storage for the on-premises application.

C. Copy the data from Amazon S3 to Amazon FSx for Windows File Server. Configure an Amazon FSx File Gateway to provide storage for the on-premises application.

D. Configure an on-premises file server. Use the Amazon S3 API to connect to S3 storage. Configure the application to access the storage from the on-premises file server.

【參考答案】B

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## 【題號】question 1004

【題目】

A company hosts its enterprise resource planning (ERP) system in the us-east-1 Region. The system runs on Amazon EC2 instances. Customers use a public API that is hosted on the EC2 instances to exchange information with the ERP system. International customers report slow API response times from their data centers.

Which solution will improve response times for the international customers MOST cost-effectively?

一家公司在 us-east-1 區域託管其企業資源規劃 (ERP) 系統。該系統在 Amazon EC2 執行個體上運作。客戶使用 EC2 執行個體上託管的公共 API 與 ERP 系統交換資訊。國際客戶報告其資料中心的 API 回應時間緩慢。

哪種解決方案能夠最具成本效益地縮短國際客戶的回應時間？

【選項】

A. Create an AWS Direct Connect connection that has a public virtual interface (VIF) to provide connectivity from each customer's data center to us-east-1. Route customer API requests by using a Direct Connect gateway to the ERP system API.

B. Set up an Amazon CloudFront distribution in front of the API. Configure the CachingOptimized managed cache policy to provide improved cache efficiency.

C.Set up AWS Global Accelerator. Configure listeners for the necessary ports. Configure endpoint groups for the appropriate Regions to distribute traffic. Create an endpoint in the group for the API.

D.Use AWS Site-to-Site VPN to establish dedicated VPN tunnels between Regions and customer networks. Route traffic to the API over the VPN connections.

【參考答案】B

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## 【題號】question 1005

【題目】

A company tracks customer satisfaction by using surveys that the company hosts on its website. The surveys sometimes reach thousands of customers every hour. Survey results are currently sent in email messages to the company so company employees can manually review results and assess customer sentiment.

The company wants to automate the customer survey process. Survey results must be available for the previous 12 months.

Which solution will meet these requirements in the MOST scalable way?

一家公司透過在其網站上進行的調查來追蹤客戶滿意度。有時每小時都會有數千名客戶接受調查。調查結果目前透過電子郵件發送給公司，以便公司員工可以手動查看結果並評估客戶情緒。

該公司希望實現客戶調查流程的自動化。必須提供過去 12 個月的調查結果。

哪種解決方案能夠以最具可擴展性的方式滿足這些要求？

【選項】

A.Send the survey results data to an Amazon API Gateway endpoint that is connected to an Amazon Simple Queue Service (Amazon SQS) queue. Create an AWS Lambda function to poll the SQS queue, call Amazon Comprehend for sentiment analysis, and save the results to an Amazon DynamoDB table. Set the TTL for all records to 365 days in the future.

B. Send the survey results data to an API that is running on an Amazon EC2 instance. Configure the API to store the survey results as a new record in an Amazon DynamoDB table, call Amazon Comprehend for sentiment analysis, and save the results in a second DynamoDB table. Set the TTL for all records to 365 days in the future.

C. Write the survey results data to an Amazon S3 bucket. Use S3 Event Notifications to invoke an AWS Lambda function to read the data and call Amazon Rekognition for sentiment analysis. Store the sentiment analysis results in a second S3 bucket. Use S3 lifecycle policies on each bucket to expire objects after 365 days.

D. Send the survey results data to an Amazon API Gateway endpoint that is connected to an Amazon Simple Queue Service (Amazon SQS) queue. Configure the SQS queue to invoke an AWS Lambda function that calls Amazon Lex for sentiment analysis and saves the results to an Amazon DynamoDB table. Set the TTL for all records to 365 days in the future.

【參考答案】A

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## 【題號】question 1006

### 【題目】

A company uses AWS Systems Manager for routine management and patching of Amazon EC2 instances. The EC2 instances are in an IP address type target group behind an Application Load Balancer (ALB).

New security protocols require the company to remove EC2 instances from service during a patch. When the company attempts to follow the security protocol during the next patch, the company receives errors during the patching window.

Which combination of solutions will resolve the errors? (Choose two.)

一家公司使用 AWS Systems Manager 對 Amazon EC2 執行個體進行日常管理和修補。EC2 執行個體位於應用程式負載平衡器 (ALB) 後方的 IP 位址類型目標群組中。

新的安全協定要求公司在修補程式期間從服務中刪除 EC2 執行個體。當公司嘗試在下一個補丁期間遵循安全協議時，公司會在補丁視窗期間收到錯誤。

哪種解決方案組合可以解決這些錯誤？（選兩個。）

**【選項】**

- A.Change the target type of the target group from IP address type to instance type.
- B.Continue to use the existing Systems Manager document without changes because it is already optimized to handle instances that are in an IP address type target group behind an ALB.
- C.Implement the AWSEC2-PatchLoadBalancerInstance Systems Manager Automation document to manage the patching process.
- D.Use Systems Manager Maintenance Windows to automatically remove the instances from service to patch the instances.
- E.Configure Systems Manager State Manager to remove the instances from service and manage the patching schedule. Use ALB health checks to re-route traffic.

**【參考答案】**CD

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**【題號】question 1007**

**【題目】**

A medical company wants to perform transformations on a large amount of clinical trial data that comes from several customers. The company must extract the data from a relational database that contains the customer data. Then the company will transform the data by using a series of complex rules. The company will load the data to Amazon S3 when the transformations are complete.

All data must be encrypted where it is processed before the company stores the data in Amazon S3. All data must be encrypted by using customer-specific keys.

Which solution will meet these requirements with the LEAST amount of operational effort?

一家醫療公司希望對來自多個客戶的大量臨床試驗數據進行轉換。公司必須從包含客戶資料的關聯式資料庫中提取資料。然後該公司將使用一系列複雜的規則來轉換資料。轉換完成後，該公司會將資料載入 Amazon S3。

在公司將資料儲存在 Amazon S3 之前，所有資料都必須在處理時進行加密。所有資料都必須使用客戶特定的金鑰進行加密。

哪種解決方案能夠以最少的營運工作量滿足這些要求？

**【選項】**

- A.Create one AWS Glue job for each customer. Attach a security configuration to each job that uses server-side encryption with Amazon S3 managed keys (SSE-S3) to encrypt the data.
- B.Create one Amazon EMR cluster for each customer. Attach a security configuration to each cluster that uses client-side encryption with a custom client-side root key (CSE-Custom) to encrypt the data.
- C.Create one AWS Glue job for each customer. Attach a security configuration to each job that uses client-side encryption with AWS KMS managed keys (CSE-KMS) to encrypt the data.
- D.Create one Amazon EMR cluster for each customer. Attach a security configuration to each cluster that uses server-side encryption with AWS KMS keys (SSE-KMS) to encrypt the data.

**【參考答案】C**

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**【題號】question 1008**

**【題目】**

A company hosts a website analytics application on a single Amazon EC2 On-Demand Instance. The analytics application is highly resilient and is designed to run in stateless mode.

The company notices that the application is showing signs of performance degradation during busy times and is presenting 5xx errors. The company needs to make the application scale seamlessly.

Which solution will meet these requirements MOST cost-effectively?

一家公司在單一 Amazon EC2 隨選執行個體上託管網站分析應用程式。此分析應用程式具有高度彈性，並且設計為在無狀態模式下運行。

該公司注意到該應用程式在繁忙時間顯示出效能下降的跡象，並出現 5xx 錯誤。公司需要使應用程式無縫擴展。

哪一種解決方案能夠最具成本效益地滿足這些要求？

**【選項】**

A.Create an Amazon Machine Image (AMI) of the web application. Use the AMI to launch a second EC2 On-Demand Instance. Use an Application Load Balancer to distribute the load across the two EC2 instances.

B.Create an Amazon Machine Image (AMI) of the web application. Use the AMI to launch a second EC2 On-Demand Instance. Use Amazon Route 53 weighted routing to distribute the load across the two EC2 instances.

C.Create an AWS Lambda function to stop the EC2 instance and change the instance type. Create an Amazon CloudWatch alarm to invoke the Lambda function when CPU utilization is more than 75%.

D.Create an Amazon Machine Image (AMI) of the web application. Apply the AMI to a launch template. Create an Auto Scaling group that includes the launch template. Configure the launch template to use a Spot Fleet. Attach an Application Load Balancer to the Auto Scaling group.

**【參考答案】D**

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**【題號】question 1009**

**【題目】**

A company runs an environment where data is stored in an Amazon S3 bucket. The objects are accessed frequently throughout the day. The company has strict data encryption requirements for data that is stored in the S3 bucket. The company currently uses AWS Key Management Service (AWS KMS) for encryption.

The company wants to optimize costs associated with encrypting S3 objects without making additional calls to AWS KMS.

Which solution will meet these requirements?

一家公司運作的環境中的資料儲存在 Amazon S3 儲存桶中。這些物品全天被頻繁訪問。該公司對儲存在S3儲存桶中的資料有嚴格的資料加密要求。該公司目前使用 AWS Key Management Service (AWS KMS) 進行加密。

該公司希望優化與加密 S3 物件相關的成本，而不需要額外呼叫 AWS KMS。

哪種解決方案可以滿足這些要求？

**【選項】**

- A. Use server-side encryption with Amazon S3 managed keys (SSE-S3).
- B. Use an S3 Bucket Key for server-side encryption with AWS KMS keys (SSE-KMS) on the new objects.
- C. Use client-side encryption with AWS KMS customer managed keys.
- D. Use server-side encryption with customer-provided keys (SSE-C) stored in AWS KMS.

**【參考答案】B**

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**【題號】question 1010**

**【題目】**

A company runs multiple workloads on virtual machines (VMs) in an on-premises data center. The company is expanding rapidly. The on-premises data center is not able to scale fast enough to meet business needs. The company wants to migrate the workloads to AWS.

The migration is time sensitive. The company wants to use a lift-and-shift strategy for non-critical workloads.

Which combination of steps will meet these requirements? (Choose three.)

一家公司在本地資料中心的虛擬機器 (VM) 上執行多個工作負載。公司正在迅速擴張。本地資料中心無法快速擴展以滿足業務需求。該公司希望將工作負載遷移到 AWS。

遷移具有時間敏感性。該公司希望對非關鍵工作負載使用直接遷移策略。

哪一種步驟組合可以滿足這些要求？（選擇三項。）

**【選項】**

- A. Use the AWS Schema Conversion Tool (AWS SCT) to collect data about the VMs.
- B. Use AWS Application Migration Service. Install the AWS Replication Agent on the VMs.
- C. Complete the initial replication of the VMs. Launch test instances to perform acceptance tests on the VMs.
- D. Stop all operations on the VMs. Launch a cutover instance.
- E. Use AWS App2Container (A2C) to collect data about the VMs.
- F. Use AWS Database Migration Service (AWS DMS) to migrate the VMs.

**【參考答案】**BCD

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**【題號】question 1011**

**【題目】**

A company hosts an application in a private subnet. The company has already integrated the application with Amazon Cognito. The company uses an Amazon Cognito user pool to authenticate users.

The company needs to modify the application so the application can securely store user documents in an Amazon S3 bucket.

Which combination of steps will securely integrate Amazon S3 with the application?  
(Choose two.)

本公司在私有子網路中託管應用程式。該公司已經將該應用程式與 Amazon Cognito 整合。該公司使用 Amazon Cognito 用戶池對用戶進行身份驗證。

該公司需要修改應用程序，以便應用程式可以將用戶文件安全地儲存在 Amazon S3 儲存桶中。

哪種步驟組合可以安全地將 Amazon S3 與應用程式整合？（選兩個。）

**【選項】**

A.Create an Amazon Cognito identity pool to generate secure Amazon S3 access tokens for users when they successfully log in.

B.Use the existing Amazon Cognito user pool to generate Amazon S3 access tokens for users when they successfully log in.

C.Create an Amazon S3 VPC endpoint in the same VPC where the company hosts the application.

D.Create a NAT gateway in the VPC where the company hosts the application. Assign a policy to the S3 bucket to deny any request that is not initiated from Amazon Cognito.

E.Attach a policy to the S3 bucket that allows access only from the users' IP addresses.

【參考答案】AC

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## 【題號】question 1012

【題目】

A company has a three-tier web application that processes orders from customers. The web tier consists of Amazon EC2 instances behind an Application Load Balancer. The processing tier consists of EC2 instances. The company decoupled the web tier and processing tier by using Amazon Simple Queue Service (Amazon SQS). The storage layer uses Amazon DynamoDB.

At peak times, some users report order processing delays and hangs. The company has noticed that during these delays, the EC2 instances are running at 100% CPU usage, and the SQS queue fills up. The peak times are variable and unpredictable.

The company needs to improve the performance of the application.

Which solution will meet these requirements?

一家公司有一個三層 Web 應用程式來處理客戶的訂單。Web 層由應用程式負載平衡器後面的 Amazon EC2 執行個體組成。處理層由 EC2 執行個體組成。該公司使用 Amazon Simple Queue Service (Amazon SQS) 將 Web 層和處理層解耦。儲存層使用 Amazon DynamoDB。

在高峰時段，一些用戶報告訂單處理延遲和大廳。該公司注意到，在這些延遲期間，EC2 執行個體的 CPU 使用率為 100%，且 SQS 佇列已滿。高峰時間是可變且不可預測的。

公司需要提高應用程式的效能。

哪種解決方案可以滿足這些要求？

**【選項】**

A. Use scheduled scaling for Amazon EC2 Auto Scaling to scale out the processing tier instances for the duration of peak usage times. Use the CPU Utilization metric to determine when to scale.

B. Use Amazon ElastiCache for Redis in front of the DynamoDB backend tier. Use target utilization as a metric to determine when to scale.

C. Add an Amazon CloudFront distribution to cache the responses for the web tier. Use HTTP latency as a metric to determine when to scale.

D. Use an Amazon EC2 Auto Scaling target tracking policy to scale out the processing tier instances. Use the ApproximateNumberOfMessages attribute to determine when to scale.

**【參考答案】D**

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**【題號】question 1013**

**【題目】**

A company's production environment consists of Amazon EC2 On-Demand Instances that run constantly between Monday and Saturday. The instances must run for only 12 hours on Sunday and cannot tolerate interruptions. The company wants to cost-optimize the production environment.

Which solution will meet these requirements MOST cost-effectively?

本公司的生產環境由週一至週六持續運作的 Amazon EC2 隨選執行個體組成。這些實例在周日只能運行 12 小時，並且不能容忍中斷。該公司希望優化生產環境的成本。

哪一種解決方案能夠最具成本效益地滿足這些要求？

**【選項】**

A. Purchase Scheduled Reserved Instances for the EC2 instances that run for only 12 hours on Sunday. Purchase Standard Reserved Instances for the EC2 instances that run constantly between Monday and Saturday.

B. Purchase Convertible Reserved Instances for the EC2 instances that run for only 12 hours on Sunday. Purchase Standard Reserved Instances for the EC2 instances that run constantly between Monday and Saturday.

C. Use Spot Instances for the EC2 instances that run for only 12 hours on Sunday. Purchase Standard Reserved Instances for the EC2 instances that run constantly between Monday and Saturday.

D. Use Spot Instances for the EC2 instances that run for only 12 hours on Sunday. Purchase Convertible Reserved Instances for the EC2 instances that run constantly between Monday and Saturday.

【參考答案】A

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## 【題號】question 1014

【題目】

A digital image processing company wants to migrate its on-premises monolithic application to the AWS Cloud. The company processes thousands of images and generates large files as part of the processing workflow.

The company needs a solution to manage the growing number of image processing jobs. The solution must also reduce the manual tasks in the image processing workflow. The company does not want to manage the underlying infrastructure of the solution.

Which solution will meet these requirements with the LEAST operational overhead?

一家數位影像處理公司希望將其本地整體應用程式遷移到 AWS 雲端。作為處理工作流程的一部分，該公司處理數千張影像並產生大型檔案。

該公司需要一個解決方案來管理不斷成長的影像處理作業。該解決方案還必須減少影像處理工作流程中的手動任務。該公司不想管理解決方案的底層基礎設施。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

**【選項】**

A. Use Amazon Elastic Container Service (Amazon ECS) with Amazon EC2 Spot Instances to process the images. Configure Amazon Simple Queue Service (Amazon SQS) to orchestrate the workflow. Store the processed files in Amazon Elastic File System (Amazon EFS).

B. Use AWS Batch jobs to process the images. Use AWS Step Functions to orchestrate the workflow. Store the processed files in an Amazon S3 bucket.

C. Use AWS Lambda functions and Amazon EC2 Spot Instances to process the images. Store the processed files in Amazon FSx.

D. Deploy a group of Amazon EC2 instances to process the images. Use AWS Step Functions to orchestrate the workflow. Store the processed files in an Amazon Elastic Block Store (Amazon EBS) volume.

**【參考答案】B**

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**【題號】question 1015**

**【題目】**

A company's image-hosting website gives users around the world the ability to upload, view, and download images from their mobile devices. The company currently hosts the static website in an Amazon S3 bucket.

Because of the website's growing popularity, the website's performance has decreased. Users have reported latency issues when they upload and download images.

The company must improve the performance of the website.

Which solution will meet these requirements with the LEAST implementation effort?

一家公司的圖像託管網站使世界各地的用戶能夠從其行動裝置上傳、檢視和下載圖像。該公司目前在 Amazon S3 儲存桶中託管靜態網站。

由於網站越來越受歡迎，網站的效能有所下降。用戶在上傳和下載圖像時報告了延遲問題。

公司必須提高網站的效能。

哪種解決方案能夠以最少的實工作來滿足這些要求？

**【選項】**

A. Configure an Amazon CloudFront distribution for the S3 bucket to improve the download performance. Enable S3 Transfer Acceleration to improve the upload performance.

B. Configure Amazon EC2 instances of the right sizes in multiple AWS Regions. Migrate the application to the EC2 instances. Use an Application Load Balancer to distribute the website traffic equally among the EC2 instances. Configure AWS Global Accelerator to address global demand with low latency.

C. Configure an Amazon CloudFront distribution that uses the S3 bucket as an origin to improve the download performance. Configure the application to use CloudFront to upload images to improve the upload performance. Create S3 buckets in multiple AWS Regions. Configure replication rules for the buckets to replicate users' data based on the users' location. Redirect downloads to the S3 bucket that is closest to each user's location.

D. Configure AWS Global Accelerator for the S3 bucket to improve network performance. Create an endpoint for the application to use Global Accelerator instead of the S3 bucket.

**【參考答案】**A

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**【題號】question 1016**

**【題目】**

A company runs an application in a private subnet behind an Application Load Balancer (ALB) in a VPC. The VPC has a NAT gateway and an internet gateway. The application calls the Amazon S3 API to store objects.

According to the company's security policy, traffic from the application must not travel across the internet.

Which solution will meet these requirements MOST cost-effectively?

一家公司在 VPC 中應用程式負載平衡器 (ALB) 後面的私有子網路中運行應用程式。VPC 具有 NAT 閘道和 Internet 閘道。應用程式呼叫 Amazon S3 API 來儲存物件。

根據公司的安全政策, 來自應用程式的流量不得透過網路傳輸。

哪一種解決方案能夠最具成本效益地滿足這些要求？

**【選項】**

A. Configure an S3 interface endpoint. Create a security group that allows outbound traffic to Amazon S3.

B. Configure an S3 gateway endpoint. Update the VPC route table to use the endpoint.

C. Configure an S3 bucket policy to allow traffic from the Elastic IP address that is assigned to the NAT gateway.

D. Create a second NAT gateway in the same subnet where the legacy application is deployed. Update the VPC route table to use the second NAT gateway.

**【參考答案】**B

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**【題號】question 1017**

**【題目】**

A company has an application that runs on an Amazon Elastic Kubernetes Service (Amazon EKS) cluster on Amazon EC2 instances. The application has a UI that uses Amazon DynamoDB and data services that use Amazon S3 as part of the application deployment.

The company must ensure that the EKS Pods for the UI can access only Amazon DynamoDB and that the EKS Pods for the data services can access only Amazon S3. The company uses AWS Identity and Access Management (IAM).

Which solution meets these requirements?

一家公司有一個在 Amazon EC2 執行個體上的 Amazon Elastic Kubernetes Service (Amazon EKS) 叢集上執行的應用程式。該應用程式具有使用 Amazon DynamoDB 的 UI 和使用 Amazon S3 作為應用程式部署一部分的資料服務。

公司必須確保 UI 的 EKS Pod 只能存取 Amazon DynamoDB, 且資料服務的 EKS Pod 只能存取 Amazon S3。該公司使用 AWS Identity and Access Management (IAM)。

哪種解決方案滿足這些要求？

**【選項】**

A.Create separate IAM policies for Amazon S3 and DynamoDB access with the required permissions. Attach both IAM policies to the EC2 instance profile. Use role-based access control (RBAC) to control access to Amazon S3 or DynamoDB for the respective EKS Pods.

B.Create separate IAM policies for Amazon S3 and DynamoDB access with the required permissions. Attach the Amazon S3 IAM policy directly to the EKS Pods for the data services and the DynamoDB policy to the EKS Pods for the UI.

C.Create separate Kubernetes service accounts for the UI and data services to assume an IAM role. Attach the AmazonS3FullAccess policy to the data services account and the AmazonDynamoDBFullAccess policy to the UI service account.

D.Create separate Kubernetes service accounts for the UI and data services to assume an IAM role. Use IAM Role for Service Accounts (IRSA) to provide access to the EKS Pods for the UI to Amazon S3 and the EKS Pods for the data services to DynamoDB.

**【參考答案】**D

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**【題號】question 1018**

**【題目】**

A company needs to give a globally distributed development team secure access to the company's AWS resources in a way that complies with security policies.

The company currently uses an on-premises Active Directory for internal authentication. The company uses AWS Organizations to manage multiple AWS accounts that support multiple projects.

The company needs a solution to integrate with the existing infrastructure to provide centralized identity management and access control.

Which solution will meet these requirements with the LEAST operational overhead?

公司需要讓分佈在全球的開發團隊以符合安全策略的方式安全地存取公司的 AWS 資源。

該公司目前使用本機 Active Directory 進行內部身份驗證。該公司使用 AWS Organizations 管理支援多個專案的多個 AWS 帳戶。

該公司需要一個解決方案來與現有基礎設施集成，以提供集中的身份管理和存取控制。

哪種解決方案能夠以最少的營運開銷滿足這些要求？

**【選項】**

A.Set up AWS Directory Service to create an AWS managed Microsoft Active Directory on AWS. Establish a trust relationship with the on-premises Active Directory. Use IAM roles that are assigned to Active Directory groups to access AWS resources within the company's AWS accounts.

B.Create an IAM user for each developer. Manually manage permissions for each IAM user based on each user's involvement with each project. Enforce multi-factor authentication (MFA) as an additional layer of security.

C.Use AD Connector in AWS Directory Service to connect to the on-premises Active Directory. Integrate AD Connector with AWS IAM Identity Center. Configure permissions sets to give each AD group access to specific AWS accounts and resources.

D.Use Amazon Cognito to deploy an identity federation solution. Integrate the identity federation solution with the on-premises Active Directory. Use Amazon Cognito to provide access tokens for developers to access AWS accounts and resources.

**【參考答案】C**

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**【題號】question 1019**

**【題目】**

A company is developing an application in the AWS Cloud. The application's HTTP API contains critical information that is published in Amazon API Gateway. The critical information must be accessible from only a limited set of trusted IP addresses that belong to the company's internal network.

Which solution will meet these requirements?

一家公司正在 AWS 雲端中開發應用程式。應用程式的 HTTP API 包含在 Amazon API Gateway 中發佈的關鍵資訊。必須只能從屬於公司內部網路的一組有限的可信任 IP 位址存取關鍵資訊。

哪種解決方案可以滿足這些要求？

**【選項】**

- A.Set up an API Gateway private integration to restrict access to a predefined set of IP addresses.
- B.Create a resource policy for the API that denies access to any IP address that is not specifically allowed.
- C.Directly deploy the API in a private subnet. Create a network ACL. Set up rules to allow the traffic from specific IP addresses.
- D.Modify the security group that is attached to API Gateway to allow inbound traffic from only the trusted IP addresses.

**【參考答案】B**

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